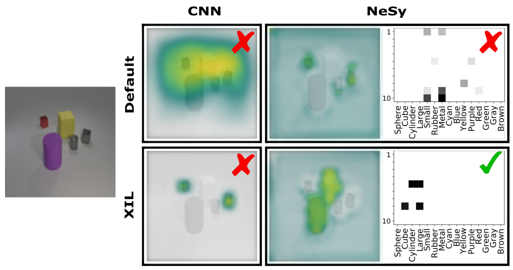

Figure 5: **Example explanations (from test set) of different model and training settings on CLEVR-Hans3.** Red crosses denote false, green checks correct predictions.

Evaluation. Apart from qualitatively investigating the explanations of the models, we used the classification accuracy on the validation and test set as an indication of a model's ability to make predictions based on correct reasons. If the accuracy is high on the confounded validation set but low on the non-confounded test set, it is fair to assume that the model focuses on the confounding factors of the data set to achieve a high validation accuracy.

6.1. Visual XIL fails on CLEVR-Hans

We first demonstrate the results of training a standard CNN for classification.

CNN produces Clever-Hans moment. As Tab. 2 indicates, the default CNN is prone to focusing on the confounding factors of the data sets. It reaches near perfect classification accuracies in the confounded validation sets but much lower accuracy in the non-confounded test sets. Interestingly, the main difficulty of the standard CNN for CLEVR-Hans3 appears to lie in the gray color confounder of class 1, whereas the confounding material of class 2 does not appear to be a difficulty for the model (*cf.* Appendix).

Examples of visual explanations of the default CNN for CLEVR-Hans3 images are presented in Fig. 5. Note these explanations appear rather unspecific and ambiguous, and it is not clear whether the model has learned the two object class rules of CLEVR-Hans3.

Revising Visual Explanations via XIL. We next apply XIL to the CNN model to improve its explanations. As in [47, 44] we set $r(A^v, \hat{e}^v)$ to the mean squared error between user annotation and model explanation. We simulate a user by providing ground-truth segmentation masks for each class relevant object in the train set. In this way, we could improve the model's explanations to focus more on the relevant objects of the scene.

An example of the revised visual explanations of the CNN with XIL can be found in Fig. 5 again visualized via Grad-CAMs. Compared to the not revised model, one can now clearly detect which objects are relevant for the model's prediction. However, the model's learned concept seems to not agree with the correct class rule, *cf.* Fig. 4, and thus, in this case, it is not able to predict the correct class. Further, it is still ambiguous what concepts about those objects are relevant for the model. The accuracies in Tab. 2 lastly indicate that correcting the visual explanations improved the overall test accuracy marginally, however comparing to the near-perfect validation accuracy, it is clear the model still focuses on confounding factors.

Model	Validation (confounded)	Test (non-confounded)
CLEVR-Hans3		
CNN (Default)	99.55 ± 0.10	70.34 ± 0.30
CNN (XIL)	99.69 ± 0.08	70.77 ± 0.37
NeSy (Default)	98.55 ± 0.27	$\circ$ $\mathbf{81.71 \pm 3.09}$
NeSy XIL	100.00 ± 0.00	$\bullet$ $\mathbf{91.31 \pm 3.13}$
CLEVR-Hans7		
CNN (Default)	96.09 ± 0.19	84.50 ± 1.04
CNN (XIL)	96.08 ± 0.25	89.26 ± 0.29
NeSy (Default)	96.88 ± 0.16	$\circ$ $\mathbf{90.97 \pm 0.91}$
NeSy XIL	98.76 ± 0.17	$\bullet$ $\mathbf{94.96 \pm 0.49}$

CLEVR-Hans3 – Global Correction Rule (¬Gray)

Model	Test (class 1)	Test (all classes)
NeSy (Default)	52.98 ± 9.60	81.71 ± 3.09
NeSy XIL	$\mathbf{83.59 \pm 8.44}$	$\mathbf{83.26 \pm 6.46}$

Table 2: **Balanced accuracies on Clevr-Hans3 and Clevr-Hans7**. The best ("•") and runner-up ("○") results are bold. We compare the test accuracy in comparison to the validation accuracy as an indication of Clever-Hans moments.

6.2. Neuro-Symbolic XIL to the Rescue

Now, we are ready to investigate how Neuro-Symbolic XIL improves upon visual XIL.

Receiving Explanations of Neuro-Symbolic model. Training the Neuro-Symbolic model in the default cross-entropy setting, we make two observations. Firstly, we can observe an increased test accuracy compared to the previous standard CNN settings. This is likely due to the class rules' relevant features now being more evident for the model to use than the standard CNN could possibly catch on to, *e.g.* the object's material. Secondly, even with a higher test accuracy than the previous model could achieve, this accuracy is still considerably below the again near perfect validation accuracy. This indicates that also the Neuro-Symbolic model is not resilient against confounding factors.

Example explanations of the Neuro-Symbolic model can be found in Fig. 5, with the symbolic explanation on the right side and the corresponding attention-based visual explanation left of this. The objects highlighted by the visual explanations depict those objects that are considered as most relevant according to the symbolic explanation (see Eq. 5 for details). These visualizations support the observation that the model also focuses on confounding factors.

Revising Neuro-Symbolic Models via Interacting with Their Explanations. We observe that the Clever-Hans moment of the Neuro-Symbolic model in the previous experiment was mainly due to errors of the reasoning module as the visual explanation correctly depicts the objects that were considered as relevant by the reasoning module. To revise the model we therefore applied XIL to the symbolic explanations via the previously used, mean-squared error regularization term. We provided the true class rules as semantic user feedback.

The resulting accuracies of the revised Neuro-Symbolic model can be found in Tab. 2 and example explanations in Fig. 5. We observe that false behaviors based on confounding factors could largely be corrected. The XIL revised Neuro-Symbolic model produces test accuracies much higher than was previously possible in all other settings, including the XIL revised CNN. To test the influence of possible Slot-Attention prediction errors we also tested revising the reasoning module when given the ground-truth symbolic representations. Indeed this way, the model could reach a near-perfect test accuracy (*cf.* Appendix).

Quantitative Analysis of Symbolic Explanations. In order to quantitatively evaluate the symbolic explanations we compute the relative L1 error on the test set between ground-truth explanations and model explanations. Briefly, for CLEVR-Hans3 NeSy XIL resulted in a reduction in L1 error compared to NeSy (Default) of: 73% (total), 64% (class 1), 76% (class 2) and 82% (class 3). For a detailed discussion *cf.* Appendix.

Revision via General Feedback Rules. Using XIL for revising a model's explanations requires that a human user interacts with the model on a sample-based level, *i.e*. the user receives a model's explanation for an individual sample and decides whether the explanation for this is acceptable or a correction on the model's explanation is necessary. This can be very tedious if a correction is not generalizable to multiple samples and must be created for each sample individually.

Consider class 1 of CLEVR-Hans3, where the confounding factor is the color gray of the large cube. Once gray has been identified as an irrelevant factor for this, but also all other classes, using NeSy XIL, a user can create a global correction rule as in Fig. 3. In other words, irrespective of the class label of a sample, the color gray should never play a role for prediction.Tab. 2(bottom) shows the test accuracies of our neuro-symbolic architecture for class 1 and, separately, over all classes. We here compare the default training mode vs. XIL with the single global correction rule. For this experiment, our explanatory loss was the RRR term [43], which has the advantage of handling negative user feedback.

As one can see, applying the correction rule has substantial advantages for class 1 test accuracies and minor advantages for the full test accuracy. These results highlight the benefit of NeSy XIL for correcting possible Clever-Hans moments via global correction rules, a previously non-trivial feature.

7. Conclusion

Neuro-Symbolic concept learners are capable of learning visual concepts by jointly understanding vision and symbolic language. However, although they combine system 1 and system 2 [15] characteristics, their complexity still makes them difficult to trust in critical applications, especially, as we have shown, if the training conditions for their system 1 component may differ from those in the test condition. However, their system 2 component allows one to identify when models are right for the wrong conceptual reasons. This allowed us to introduce the first Neuro-Symbolic Explanatory Interactive Learning approach, regularizing a model by examining and selectively penalizing its Neuro-Symbolic explanations. Our results on a newly compiled confounded benchmark data set, called CLEVR-Hans, demonstrated that semantic explanations, *i.e.*, compositional explanations at a per-object, symbolic level, can identify confounders that are not identifiable using "visual" explanations only. More importantly, feedback on this semantic level makes it possible to revise the model from focusing on these confounding factors.

Our results show that Neuro-Symbolic explanations and interactions merit further investigation. Using a semantic loss [56] would allow one to stay at the conceptual level directly. Furthermore, one should integrate a neural semantic parsing system that helps to interactively learn a joint symbolic language between the machine and the human user through decomposition [16]. Lastly, language-guided XIL [34] is an interesting approach for more natural supervision. These approaches would help to move from XIL to conversational XIL. Applying Neuro-Symbolic prior knowledge to a model may provide additional benefits to a XIL setting. Finally, it is very interesting to explore more expressive reasoning components and investigate how they help combat even more complex Clever-Hans moments. Concerning our data set, an interesting next step would be to create a confounded causal data set in the approach of [10].

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